

HAOLIN WANG

Ph.D. Student in Civil Engineering and M.S. in Computer Science and Engineering

Georgia Institute of Technology

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EDUCATION

Georgia Institute of Technology — School of Civil Engineering

Aug 2022 – Present

Ph.D. Student in Civil Engineering; M.S. in Computer Science & Engineering

- Advisor: Dr. Yi-chang (James) Tsai
- GPA: 4.00

Tongji University — School of Civil Engineering

Sept 2017 – June 2021

Bachelor of Engineering in Civil Engineering

- GPA: 3.89
- Honored as Excellent Graduate of Tongji University, 2021

RESEARCH & PROJECT EXPERIENCE

Pavement Surface Cracking Digital Twin — Thesis

Jan 2023 – Present

Advisor: Dr. Yi-chang (James) Tsai

- Studying and exploring the complete process of developing a digital twin for pavement surface cracking and implementing it in actual pavement condition assessment.

NCHRP 01-57 B: Validating Proposed Definitions for Comparable Pavement Cracking Data — Core Member

Apr 2023 – Present

PI: Dr. Yi-chang (James) Tsai

- Developing and validating comparable cracking protocols that could fundamentally represent cracking data while supporting various agencies' data aggregation and reporting needs.

Pavement Raveling Deterioration Analysis — Core Member

May – Sept 2023

Advisor: Dr. Yi-chang (James) Tsai

- Studied raveling deterioration using both traditional qualification and emerging quantitative methods to inform treatment decisions.

Multi-Timestamp Pavement Image Registration — Core Member

Jan 2023 – Present

Advisor: Dr. Yi-chang (James) Tsai

- Developed and evaluated image registration methods for multi-timestamp pavement images to eliminate offsets and enable reliable time-series analysis.

Life-Cycle Maturity Model of Intelligent Construction — Core Member

Dec 2019 – Apr 2021

Advisor: Dr. Yujie Lu

- Conducted literature review, questionnaire surveys, and hierarchical analysis of intelligent construction in urban rail transit.
- Built a maturity model recognized as an excellent project by the supervising professor.

Crack Detection in High-Speed Railway Ballastless Track Slabs — Core Member

Sept 2019 – Sept 2020

Advisor: Dr. Yu Zhou

- Developed a non-destructive, real-time assessment method for track slabs using acoustic emission technology.
- Applied finite element simulation and numerical calculation to quantify the relationship between acoustic signals and crack propagation; resulted in two granted patents.

“Mini Traffic” Urban Road Simulation Software — Team Leader

Jan – Mar 2019

Advisor: Dr. Keping Yuan

- Built an urban traffic simulation system in Python, C#, and SQL with interactive data visualization; won 3rd Prize at Shanghai University Student Computer Application Ability Competition.

PUBLICATIONS

Wang, H., & Tsai, Y.-C. (2025). Toward reliable pavement digital twins: Multi-scale spatiotemporal monitoring challenges and solutions. Proceedings of the 1st ACM SIGSPATIAL International Workshop on Spatial Digital Twins.

Yang, Z., Mohammadi, M., **Wang, H.**, & Tsai, Y.-C. (2025). A feature-based pavement image registration method for precise pavement deterioration monitoring. Computer-Aided Civil and Infrastructure Engineering, 40(10), 1276–1294. IF 9.1, SCI Q1.

Zhang, X., **Wang, H.**, Hsieh, Y. A., Yang, Z., Yezzi, A., & Tsai, Y.-C. (2025). Deep learning for crack detection: A review of learning paradigms, generalizability, and datasets. arXiv preprint arXiv:2508.10256.

Zhang, X., **Wang, H.**, & Yang, Z. (2026). Towards successful implementation of automated raveling detection: Effects of training data size, illumination difference, and spatial shift. arXiv preprint arXiv:2604.13322.

CONFERENCE PRESENTATIONS

Three NCHRP 01-57 B-related Presentations

Apr 2026

2026 Annual Road Profile Users' Group (RPUG) Conference

Cracking Data Quality Assessment Using Ground References: Developing a Standard Methodology

Apr 2025

2025 Annual Road Profile Users' Group (RPUG) Conference

Standardizing Cracking Data Quality Assessment and Quantification: Case Studies in Florida, Mississippi, and Texas

Jan 2026

105th Transportation Research Board (TRB) Annual Meeting

Impact Analysis of Image Offsets on Time-Series Crack Analysis Variability Using 3D Pavement Imaging Data

Jan 2025

104th Transportation Research Board (TRB) Annual Meeting

Multi-Year Raveling Deterioration Analysis Using Macrotexture and Aggregate Loss Indicators from 3D Pavement Surface Data

Jan 2024

103rd Transportation Research Board (TRB) Annual Meeting

Innovation of Tunnel Anchor Technology

May 2019

5th Annual International Conference for Students

TEACHING ASSISTANT EXPERIENCE

Vertically Integrated Projects (VIP)

Spring 2023 – Present

- Mentored students on crack definition standardization, spatiotemporal pavement analysis using 3D surface data, LiDAR-aided sidewalk assessment, and pavement image registration methods.

CEE-4803: AI for Smart Cities

Spring 2024 & Spring 2025

- Supported lecture and assignment preparation, grading, and lab instruction covering AI applications implemented in Python.

CEE-8813: Data Analytics for Transportation Safety

Fall 2024

- Developed and graded Python/GIS crash-analytics assignments; mentored student term-project teams.

CEE-4803: Data Analytics in Civil & Environmental Engineering

Fall 2023 & Fall 2024

- Designed assignments and supported lectures on data analysis methods for civil and environmental engineering applications.

PROFESSIONAL EXPERIENCE

Georgia Institute of Technology — Smart City Infrastructure Group

Jan 2023 – Present

Graduate Research Assistant | Principal Investigator: Dr. Yi-chang (James) Tsai

- Contributed to projects funded by the National Cooperative Highway Research Program (NCHRP) and the Georgia Department of Transportation (GDOT).
- Developed expertise in pavement asset management, time-series analysis with advanced sensing technology, and machine learning.

Tongji University — Rail Transportation Construction & Management Group

Sept 2021 – May 2022

Research Intern | Principal Investigator: Dr. Longcai Yang

- Assisted with research projects and paper writing; applied MATLAB for data analysis and technical reporting.

Kunming Survey, Design & Research Institute | China Railway Eryuan Engineering Group

Jul – Aug 2020

Design Assistant

- Used Midas Civil engineering software for structural design; authored two project reports.

PATENTS

CNIPA Patent No. CN202010987184.0. A Nondestructive Monitoring Method for Crack of Ballastless Track Slab. Sept 2020.

CNIPA Patent No. CN202010985244.5. A Nondestructive Testing Method for Ballastless Track Slab in Rail Transit. Sept 2020.

HONORS & AWARDS

- CEE Bill Schutz Graduate Teaching Assistant Award, Georgia Tech (2025)
- Excellent Graduate, Tongji University (2021)
- 1st Prize, Tongji Scholarship for Excellent Student (2020)
- Bronze, The Fifth "Excellence Cup" Competition, Tongji University (2020)
- Summer School Excellent Student, School of Transportation Engineering, Tongji University (2020)
- Excellent Student, Tongji University (2019–2020)
- 3rd Place, Shanghai University Student Computer Application Ability Competition (2019)
- Excellent Student Leader, Tongji University (2018–2019)
- Tongji Scholarship for Social Practice (2018)

SKILLS & ADDITIONAL INFORMATION

Programming & Tools: Python, ArcGIS, SQL, AutoCAD, MATLAB, Excel, C++, C#

Languages: English (TOEFL 102); Mandarin Chinese (native)

Standardized Tests: GRE 324